SevaForge

SEVA FIRST INNOVATION CHALLENGE 2026 · MAHARASHTRA ZONE

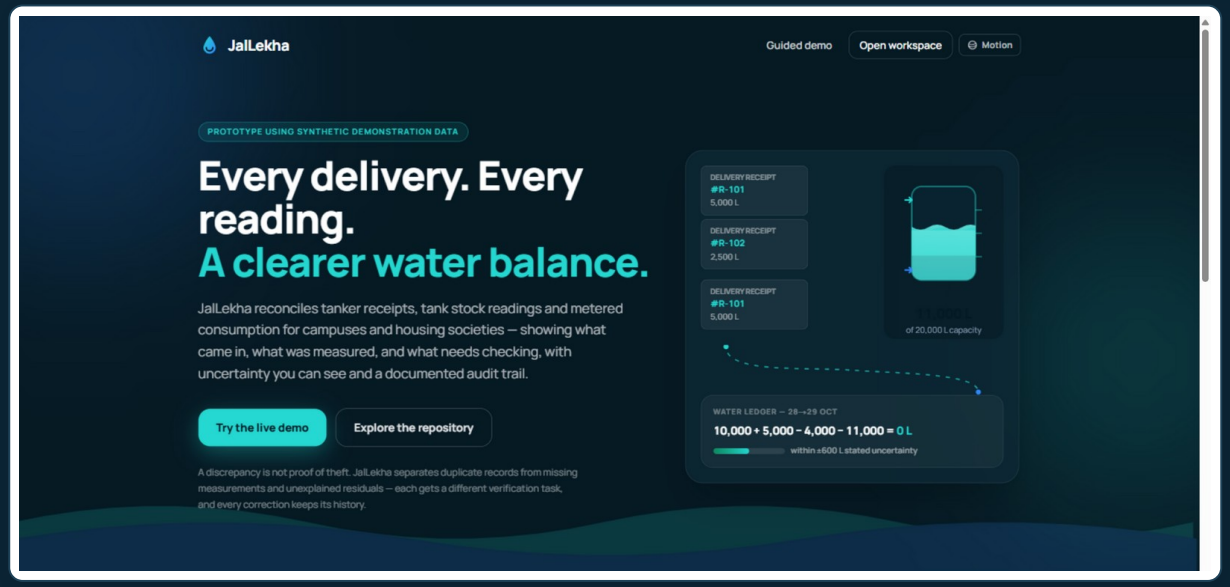
JalLekha

Every delivery. Every reading.
A clearer water balance.

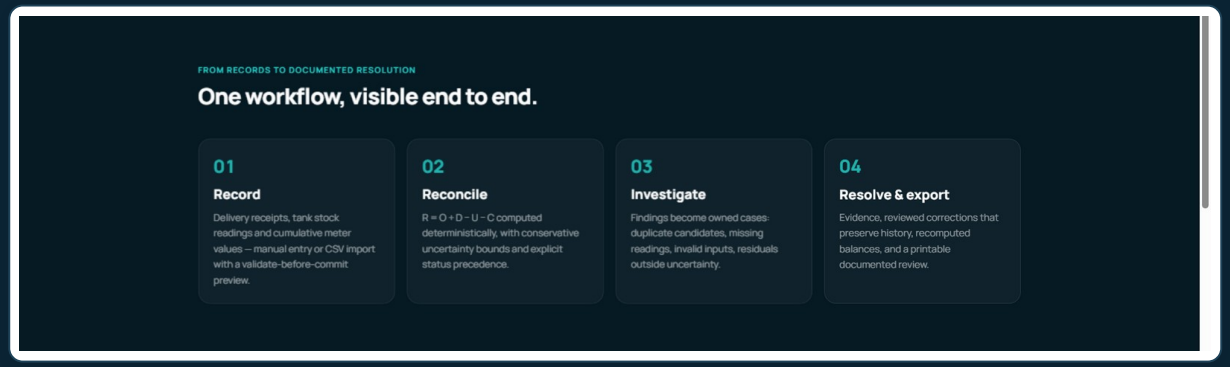
A deterministic water-balance engine that reconciles tanker receipts, tank stock readings and meter logs for campuses and housing societies - showing what came in, what was measured, and what needs checking, with uncertainty you can see and an audit trail you can defend.

Working prototype · live demo | **41 unit + 33 e2e tests green**

Sector: Sustainability - Energy, Climate and Others
Sub-theme: Technology for pollution mitigation, water conservation and waste management
Track A - Product / Technology Solution · Open Level



The live product - jallekha-na41.zai-ai.site/demo



One workflow, visible end to end - record → reconcile → investigate → resolve

THE PROBLEM

Three registers. No single water balance.

Tanker receipts

`#R-101 · 5,000 L`

paper slips + a register

Tank dipstick readings

`11,000 L ± 200`

twice a day, by hand

Meter logs

`1,00,000 → 1,04,000`

cumulative m^3, one line a day

Water doesn't add up - so staff trace entries by hand

Duplicate? Missing reading? Real loss? The register cannot say.

WHO FEELS IT

- **Hostel & campus teams**
- **Housing societies**
- **Students & residents**

WHY PAPER CANNOT CARRY THIS WORKFLOW

1. ### Same symptom, three different causes

   A duplicate receipt needs a correction, a missing reading needs a measurement, an unexplained residual needs investigation. Paper cannot tell them apart - so staff guess.

2. ### Exact-looking numbers hide uncertainty

   A dipstick is ±100-200 L, a meter ±100 L. Registers make estimates look exact: small errors trigger panic while real losses hide inside tolerance.

3. ### Corrections destroy the evidence

   Overwrite one register line and the audit trail is gone. Disputes with suppliers cannot be evidenced - so nothing gets fixed and water stays unaccounted.

THE INNOVATION

One equation. Three honest outcomes.

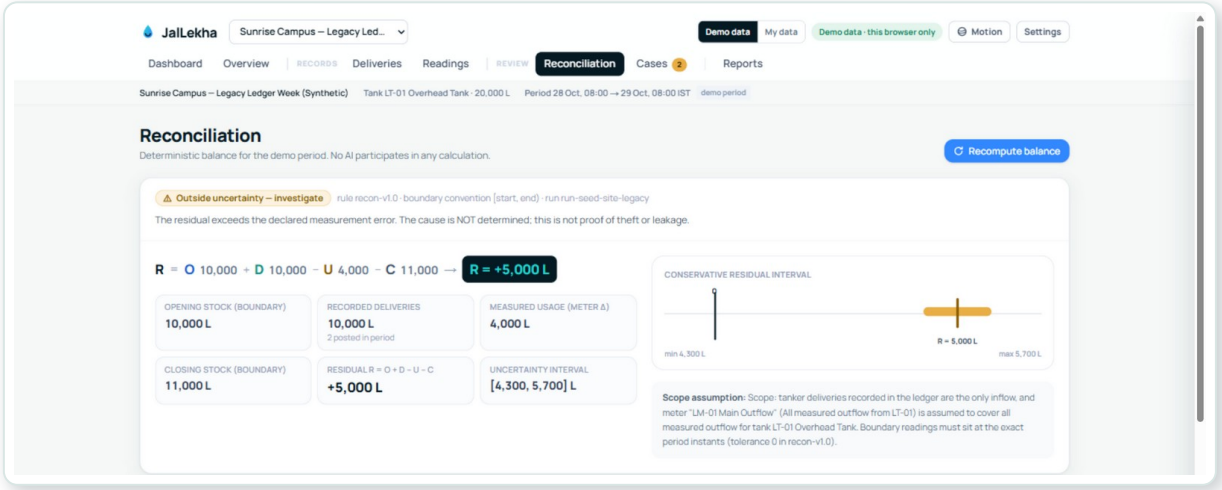
THE WATER BALANCE - recon-v1.0

$$R = O + D - U - C$$

OPENING DELIVERIES USAGE CLOSING

```
R_min = O_min + D_min - U_max - C_max
conservative summed bounds - never root-sum-square optimism
```

Status precedence: invalid → incomplete (no residual asserted) → uncertainty not quantified → within uncertainty → discrepancy



Live reconciliation - Scenario B: a double-entered receipt R-101 yields R = +5,000 L, bounds [4,300, 5,700] L, duplicate candidate linked to both rows.

SCENARIO B · DUPLICATE

Record correction

Reviewer voids the repeated receipt with a reason - R returns to 0 L and the original records stay in history.

SCENARIO C · MISSING READING

No residual invented

No closing reading? The run is 'incomplete' and asserts no residual at all. The fix is a measurement, not a guess.

SCENARIO D · UNEXPLAINED

Investigation, never accusation

R = +2,000 L in [1,400, 2,600] L opens an owned case - the cause is never auto-labelled theft or leakage.

Every number carries its uncertainty - and a discrepancy is never labelled theft or leakage.

THE ENGINEERING

A pure, tested core. AI assists - never decides.

INTERFACE - Next.js 16 · TypeScript · Tailwind

landing · guided demo · workspace: dashboard, ledger, readings, reconciliation, cases, reports

↓

DOMAIN - recon-v1.0 rule engine · pure TypeScript

balance · findings · CSV import · audit state · strict units - no framework imports, fully unit-tested

↓

DATA - DataAdapter port

browser-local demo (disclosed) · in-memory tests · shared backend (Supabase + RLS) designed as drop-in

Key decisions: decimal-safe litres (NaN / negative / over-capacity rejected, never clamped) · exact boundary instants (tolerance 0) · content-derived stable finding IDs

BUILT & VERIFIED - DATED RUN LOG, 31 OCT 2026

✓ **41 / 41 domain tests · 33 / 33 browser e2e**

Vitest + Playwright on the production build; strict tsc 0 errors, lint 0/0 - docs/VALIDATION.md

✓ **Deterministic engine - zero AI in the math**

Same inputs always give the same verdict; four signature scenarios asserted to the litre

✓ **CSV import you can trust**

Validate-before-commit preview; replaying the same file never doubles a total

✓ **Every correction keeps its history**

Audit trail + superseded runs - reviewers see what changed and why

✓ **Optional AI, bring your own key**

Schema-validated explanations & note-drafting with rule-based fallback - stored only in your browser

NOW Single-device pilot - browser-local workspace, operator CSVs, zero hardware → **NEXT** Shared authenticated workspaces - the adapter port is already designed for it → **THEN** Multi-site rollup - one campus chain or a municipal ward, same equation

THE IMPACT

Measured water is manageable water.

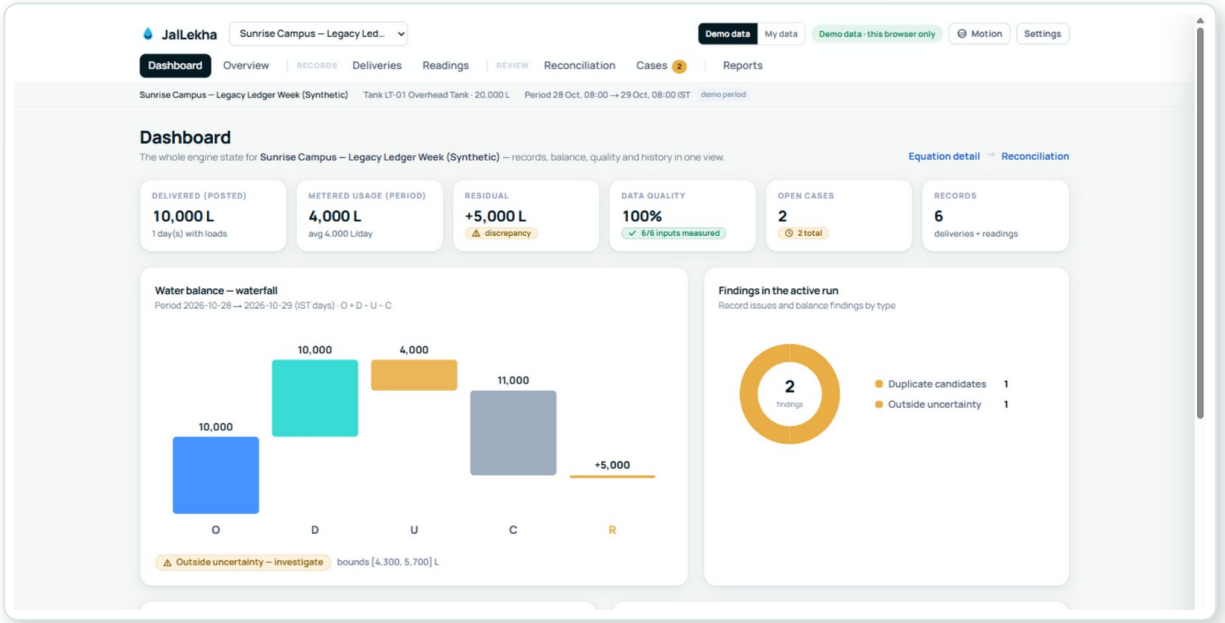
WHAT THE PILOT WILL MEASURE

01 **Planted-discrepancy detection rate**
% of seeded duplicate / missing records caught on known test ledgers

02 **Reviewer time-to-diagnosis**
minutes to the correct next action, vs tracing paper registers by hand

03 **Uncertainty coverage**
share of reconciliation periods reported with quantified bounds

04 **Case closure quality**
duplicates voided with reasons; missing readings re-measured and closed



Dashboard - balance waterfall, findings mix, meter trend, tank gauge and data quality in one view.

Baselines are set during the pilot - no litres-saved claims are made before field data exists.

CONSERVATION losses surface early; over-ordering and unnoticed leaks become owned cases

POLLUTION & HAULAGE documented demand means fewer emergency tanker trips and less idle haulage

WASTE & TRUST evidence-backed records end billing disputes and repeat reconciliation work

Four builders. One honest ask.

AJ — **Arya Jannawar**
Team Lead · product & reconciliation domain

AV — **Avani Jannawar**
Engineering & data integrity

SP — **Siddhant Santosh Pardeshi**
Quality & user workflow

VS — **Vijay Sahu**
Research & validation

WHAT WE ARE ASKING FOR

- **One pilot campus or housing society**
  4-6 weeks, one device, operator-supplied CSVs - we instrument the four impact metrics and publish the baselines
- **Mentorship & lab access**
  study design for the reviewer-time measurement, and measurement-quality practice from domain experts
- **A route to shared deployment**
  introductions toward authenticated, multi-user workspaces for a full institution - the adapter design is ready

> **"Know what came in, what was measured, and what needs checking."**

JalLekha - from Team SevaForge

github.com/Avani376/jallekha · live demo: jallekha-na41.zai-ai.site/demo